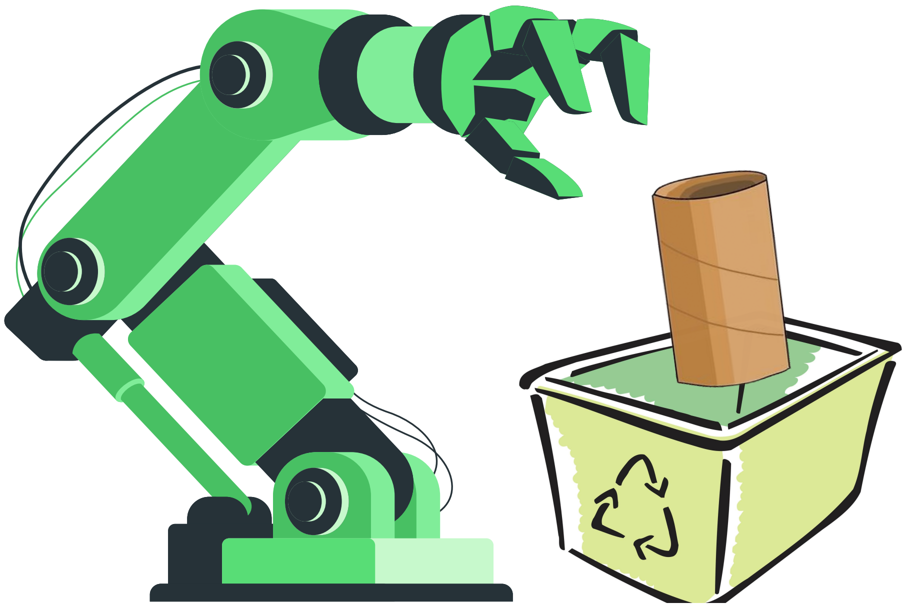
# GreenArm

EECS 4421/5324

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# Problem Background


- Public and commercial washrooms produce large amounts of toilet-related waste (tissues, crumpled paper, hygiene items).
- These are frequently thrown in the wrong bins → contaminating recycling and increasing janitorial workload.
- Unhygienic for workers who manually handle germ-contaminated waste.
- Incorrect disposal causes plumbing blockages, overflows, and higher maintenance costs.
- There is a clear need for a safer, hygienic, and automated system to manage this waste efficiently

# Our Solution

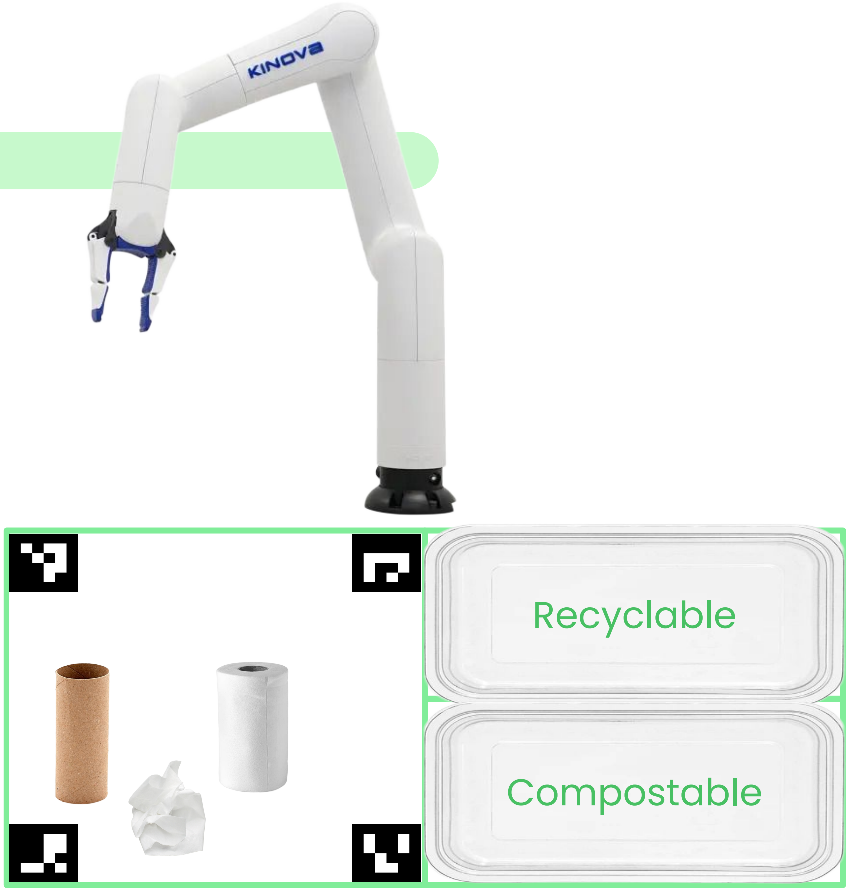
- We built Green Arm (Kinova) powered by vision.

**What’s that ?**

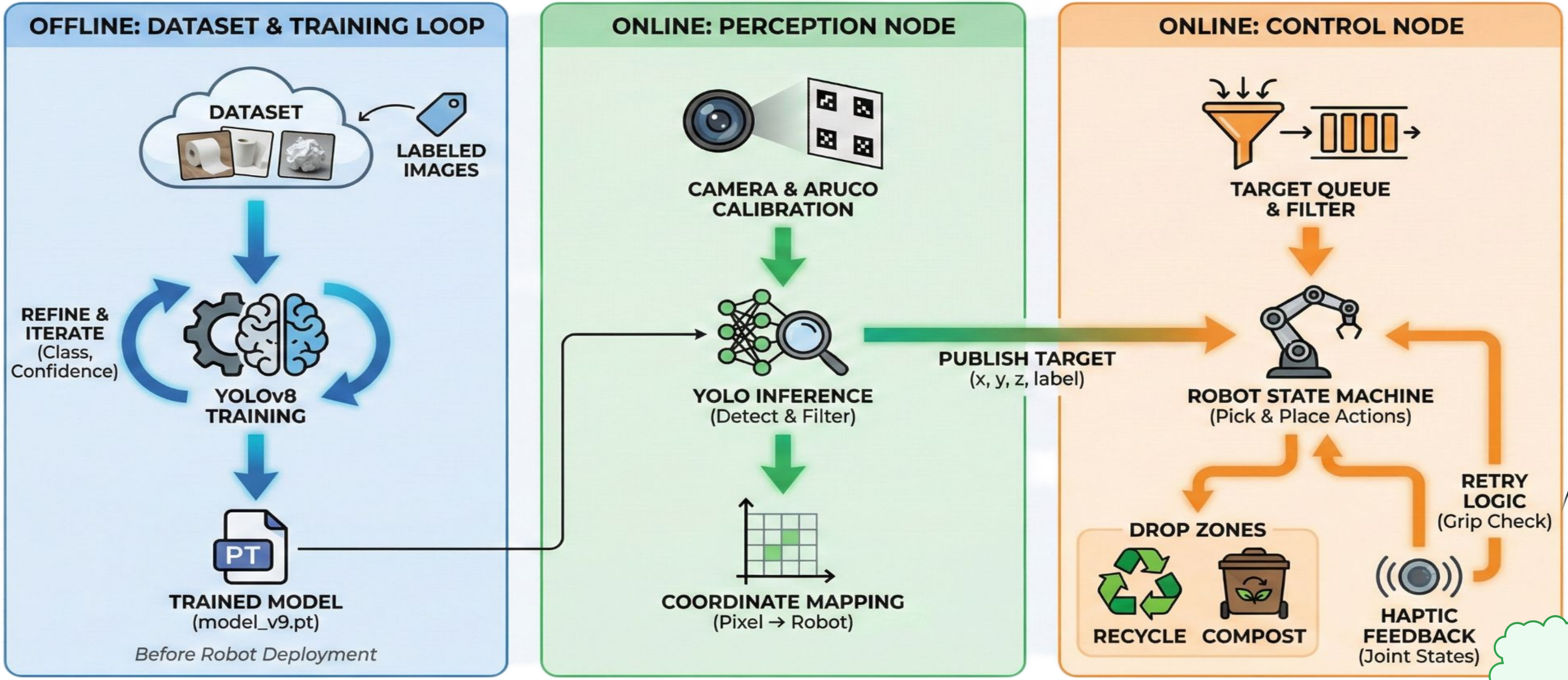
- Kinova arm powered robot that uses webcam for the vision (Object Detection and classification) and drops the waste to its appropriate destination based on the classification report.

# Our Design

- Workspace has two zones, Source and Destination. The destination zone is further divided into two sub-zones, Recycle and Compost.
- One kinova arm centered outside the workspace and camera mounted overhead, the source zone is being calibrated using Aruco markers and there are two trays onto Destination zone.



# Our Architecture



# Future Work

- More object classes & more varied waste items.
- Adaptive height and orientation pickups based on advance vision algorithm.
- Dynamic scheduling for multiple objects detected in the workspace.
- Train on more diverse and volume of data for better object detection and classification.

# Thank You !!

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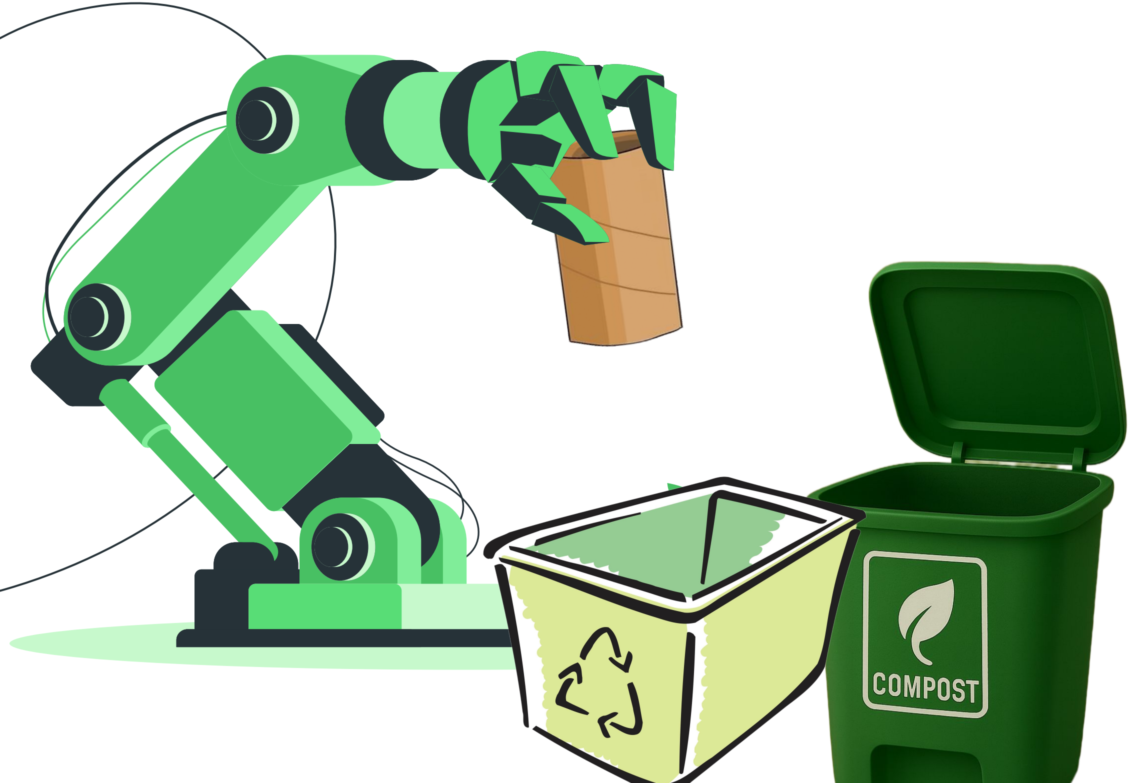


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